CPU DRAM Bandwidth Characterization

Snapdragon 8 Elite (Oryon V2) · Samsung Galaxy S25+

- Random pointer-chase (`bwprobe`) + streaming matvec (`matvec`)
- PMU calibration, cache boundary detection, latency stratification
- DRAM amplification source isolation (`prefault_exp`, `llcc_size`)
- Android 15, kernel 6.6, cpu6 pinned, freq locked to 4,473,600 kHz

Platform & Cache Hierarchy

Level	Nominal	Empirical	Notes
L1D	~96 KB/core	—	Per core
L2	12 MB (Prime)	cache ≤ 24 MB	Prime cluster shared
LLCC/SLC	"12 MB/slice"	**~112 MB effective**	Multiple slices; all flush sizes agree
DRAM	LPDDR5X 128-bit	lat **~5.5 ns/hop**	All IPs share

- bwmon-llcc-prime: hardware DRAM monitor for cpu6–7 (~4 ms samples)
- `exclude_kernel=0` mandatory — cache refills attributed to kernel on Oryon V2

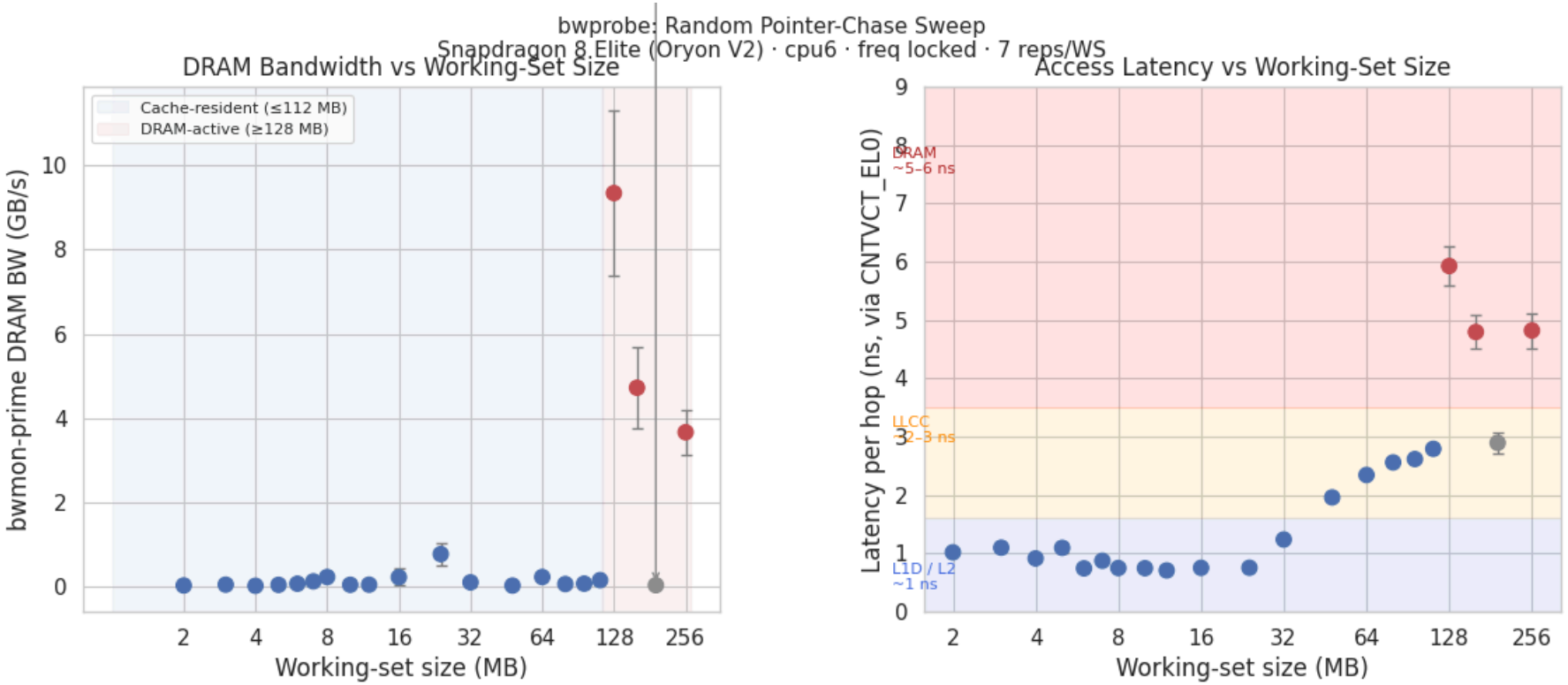
Measurement Architecture

```
Per rep: flush_caches(24 MB) → warmup (0.5 s) → chase_latency_ns (CNTVCT)
         → [bwmon snap] → 3–5 s workload → [bwmon snap]
```

Metric	Source	What it measures
`bwmon-prime`	HW tracepoint (`dcvs/bw_hwmon_meas`)	Cluster DRAM BW ground truth
`bus_access` (0x0019)	ARM PMU	Per-process DRAM transactions
`lat_ns`	CNTVCT_EL0 (no syscall)	Per-hop pointer-chase latency
`chase_GBs`	wall clock	Effective pointer-chase throughput

- `bus_access` at WS ≥ 128 MB: **1.034 ± 0.008×** vs bwmon (21 rep-points)
- `l2d_cache_refill`, `l2d_lmiss_rd`, `l3d_alloc`, LLCC PMU: all broken/inaccessible

bwprobe: Results



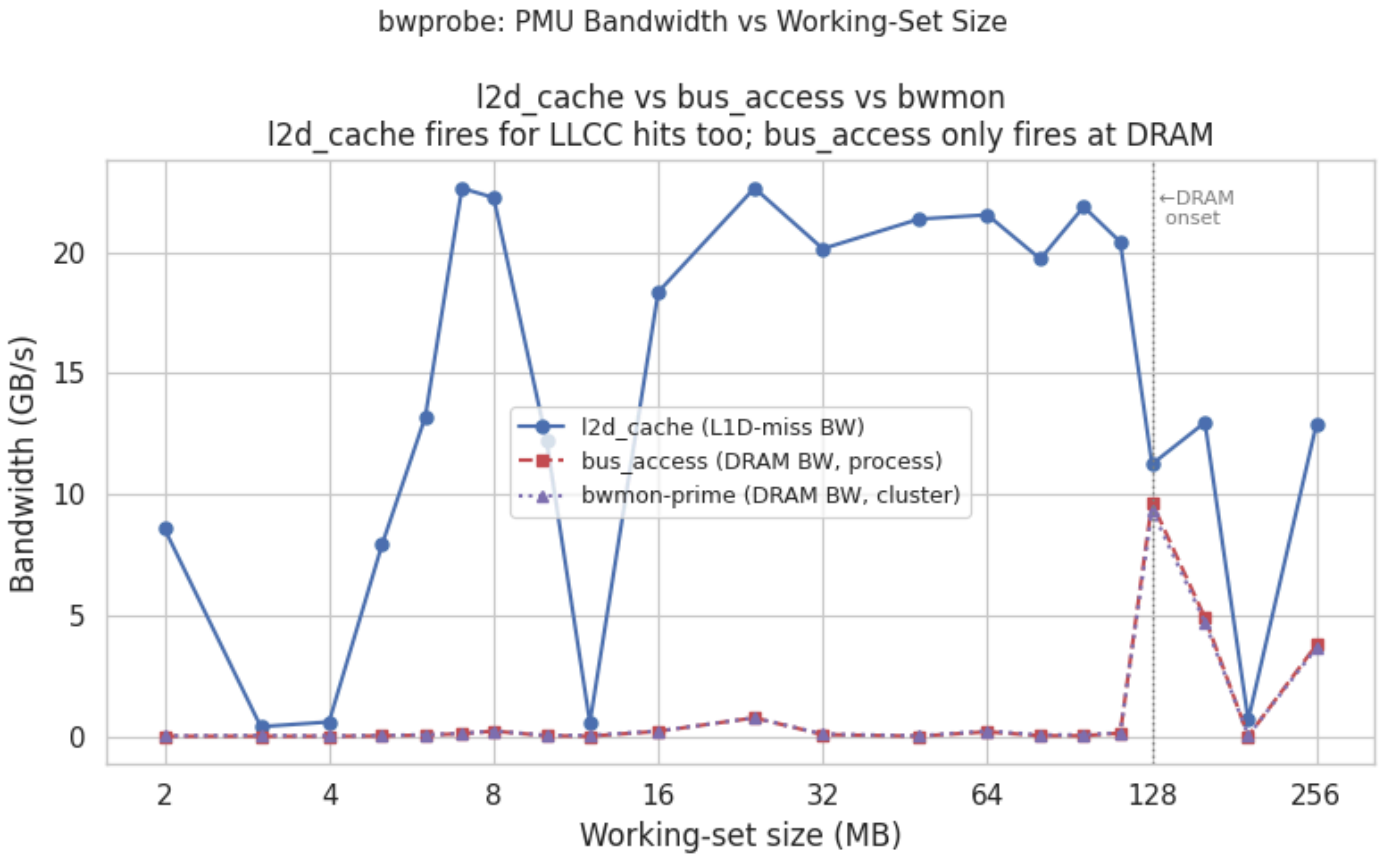
- chase_GBs and lat_ns are the **reliable regime indicators** (bwmon noisy in cache regime)
- Both show clean 2× transition at WS = 128 MB

bwprobe: Three-Tier Latency

WS	lat (ns)	chase (GB/s)	bwmon (MB/s)	Level
2–12 MB	0.7–1.1	22–23	20–228	L1D / L2
32–112 MB	1.2–2.8	19–22	25–772	LLCC
128 MB	**5.93**	**10.6**	**9,343**	**DRAM**
160–256 MB	4.8–5.9	12–13	3,600–4,700	DRAM
192 MB	2.9	21.3	34	ZRAM anomaly

- CNTVCT latency (no syscall overhead): clean 5–6× jump L2→DRAM
- 192 MB anomaly: Android ZRAM compresses buffer pages between reps → collapses to cache-resident

PMU Calibration: bus_access



- `bus_access × 64B / elapsed ≈ bwmon` at WS ≥ 128 MB: **ratio 1.034 ± 0.008**
- In cache regime: `bus_access ≈ background` (~10–228 MB/s) — **correct signal** (no per-process DRAM)
- `l2d_cache`: fires on every L1D miss; use for LLCC contention (not DRAM only)

Metric Selection

Need	Use
Per-process DRAM BW (WS ≥ 128 MB, freq locked)	`bus_access × 64 / elapsed` (1.034× accurate)
Cluster DRAM ground truth	`bwmon-llcc-prime` / `bwmon-llcc-gold`
Cache-miss pressure (any level)	`l2d_cache × 64 / elapsed`
Per-process DRAM, variable freq	`bwmon` (cluster-level only, no per-process)

Cross-processor contention characterization:

- L1D hit: `l2d_cache = 0` → zero contention ✓
- LLCC hit: `l2d_cache > 0`, `bus_access ≈ 0` → LLCC contention, no DRAM
- DRAM hit: both `l2d_cache > 0` and `bus_access > 0` → full contention

llcc_size: Effective Cache Capacity

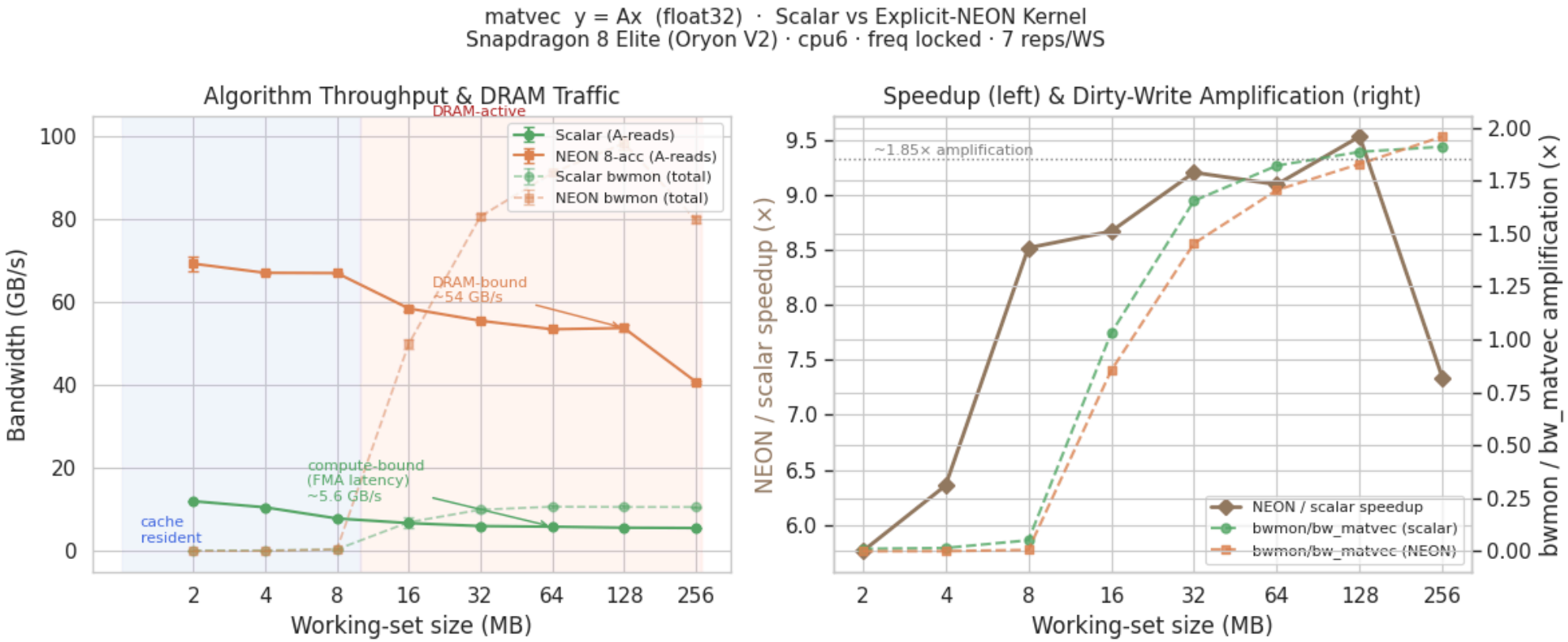
Pointer-chase with flush ∈ {24, 128, 256} MB before each rep. `lat_ns` from CNTVCT (reliable).

WS (MB)	flush=24 lat	flush=128 lat	flush=256 lat	Regime
32	1.68†	1.23	1.24	cache
64	4.92†	2.31	2.30	cache
96	2.65	2.63	2.62	cache
112	2.72	2.73	2.74	cache (boundary)
128	**5.35**	**5.37**	**5.50**	**DRAM**

†flush=24 artifact: small scrub competing with WS in LLCC during warmup.

Finding: all three flush sizes agree — boundary at WS 112→128 MB.
→ LLCC capacity ≈ **112 MB** (≈ 9× the nominal "12 MB/slice" spec)

matvec: Scalar vs NEON



- M=4096 (y=16 KB L1D-resident); x ≤ 64 KB L1D-resident; **A is sole DRAM pressure**
- Scalar: FMA-latency limited. NEON (8 accumulators): DRAM-bandwidth limited at WS=128 MB

matvec: Why BW Decreases with WS

Scalar: `s += row[j]*x[j]` → clang emits **4-wide NEON FMLA, 1 accumulator** (dep chain on `s`)

WS	N	dep chain / row	OOO parallelism	BW
2 MB	128	32 iters × 11c = 352c	2–3 rows in flight	~12 GB/s
128 MB	8192	2048 × 11c = 22,528c	1 row fills OOO window	~5.6 GB/s

- FMA ceiling: `16B × 4.47 GHz / 11 cycles = 6.5 GB/s` ≈ 5.6 GB/s ✓
- **Short rows → OOO extracts row-level ILP → higher effective accumulator count**
- NEON (8-acc): breaks dep chain → **54 GB/s** at WS=128 MB (DRAM-bandwidth bound)

matvec: DRAM Amplification

`bwmon ≈ 1.88–1.94× bw_matvec` at WS ≥ 128 MB. `prefault_exp.c` isolates the cause.

Variant	bwmon	bw_alg	amp
baseline (flush + write-init)	11,676 MB/s	6,005 MB/s	**1.94×**
`MADV_DONTNEED` on A	43 MB/s	5,996 MB/s	0.007× (confounded*)
`MADV_RANDOM` on A	11,547 MB/s	6,005 MB/s	**1.92×**

*DONTNEED collapses physical pages → zero-page → L1D cached. Not a valid clean-page test.

Conclusion: amplification is from Oryon V2 hardware stream prefetcher.

- `MADV_RANDOM` = OS readahead only; CPU hardware prefetcher is architectural, always active
- Prefetcher fetches lines ahead; A >> LLCC → lines evicted before demand → 2× DRAM per line
- Alternative: LPDDR5X 128B burst granularity (indistinguishable from userspace)
- Disabling HW prefetcher requires `CPUACTLR_EL1` (EL1 privilege only)

icc_set_bw: Software Control Loop

`icc_set_bw` tracepoint fires when DCVS updates interconnect bandwidth votes.

ICC Path	avg_bw	Meaning
`chm_apps → qns_llcc`	~81 GB/s	CPU cluster → LLCC BW vote
`llcc_mc → ebi`	~57 GB/s	LLCC → DRAM controller BW vote

```
bwmon (~4 ms)  →  DCVS reads  →  ICC vote (~40 ms)  →  DRAM freq/QoS
```

- `icc_set_bw` is aggregate system-wide — not per-process attribution
- Use `bus_access` for per-process; `bwmon` for cluster-level ground truth

Key Findings

1. **DRAM onset: WS = 128 MB** — confirmed by chase_GBs, lat_ns, bwmon; independent of flush size
2. **Effective LLCC ≈ 112 MB** — "12 MB LLCC" = one slice; cpu6 accesses ~9 slices
3. **bus_access: 1.034 ± 0.008×** — accurate at WS ≥ 128 MB; zero in cache regime = correct signal
4. **Scalar: ~5.6 GB/s** — FMA-latency bound (11 cycles); BW drops with WS = OOO row-ILP loss
5. **NEON: ~54 GB/s, 9.5× speedup** — DRAM-bandwidth bound; breaks FMA dep chain
6. **~1.94× DRAM amplification** — HW stream prefetcher; MADV_RANDOM has no effect
7. **icc_set_bw** — DCVS software loop: bwmon (4 ms) → ICC vote (40 ms) → DRAM policy

Open Questions

Question	Approach
Confirm prefetcher as amplification cause	Write `CPUACTLR_EL1` (need kernel module)
Proper clean-page test	File-backed mmap (not `MADV_DONTNEED`)
Multi-processor LLCC contention	CPU + GPU simultaneous; measure `l2d_cache` + bwmon
WS=192 MB ZRAM anomaly	Use `mlock()` to prevent page compaction
LLCC PMU access	Snapdragon Profiler via proprietary driver